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# Six ways numbers grow

The examples are teaching numbers. An input is the number you start with. An output is what the rule gives back. The tables use different units, so the lesson is the pattern, not a claim that miles and square feet are the same kind of thing. Each drawing uses its own unit and its own scale. A tall mark in one drawing is not a larger amount than a short mark in another.

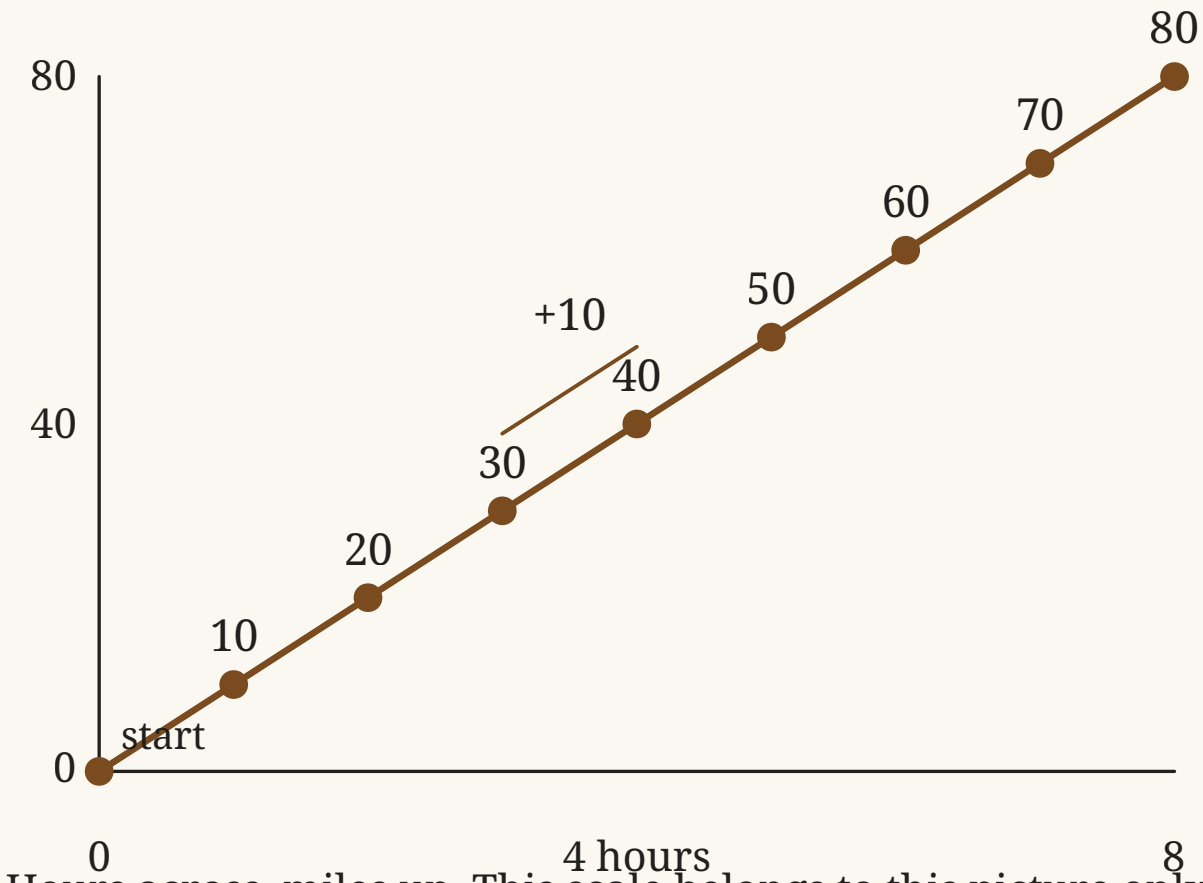
The wind lesson is one case of the square. This page sets that case beside five others.

## Linear: add the same amount

Linear growth adds the same amount for every equal step. An odometer that gains 10 miles each hour, while the vehicle holds 10 miles per hour, is the picture.

# Each hour adds 10 miles

Same addition every step. The line is straight.




Hours across, miles up. This scale belongs to this picture only.

Each hour adds 10 miles. The miles are 10, 20, 30, 40, 50, 60, 70, and 80. Each step adds 10, so the line is straight. The scale is miles, for this picture only.

| Hours | Miles |
|---|---|
| 1 | 10 |
| 2 | 20 |
| 3 | 30 |
| 4 | 40 |
| 5 | 50 |
| 6 | 60 |
| 7 | 70 |
| 8 | 80 |

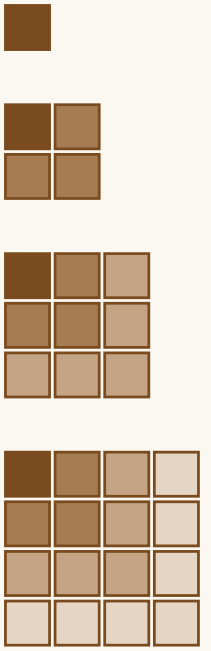
Each step adds 10. The step-to-step difference does not grow.

## Quadratic: the square

Quadratic growth follows the area of a square as the side gets longer. A square foot is a square one foot on each side. A bed 2 feet on a side covers 2 times 2, which is 4 square feet. The added pieces are 3, then 5, then 7, and so on. They grow.

# The square grows by odd pieces

Side times side. The added piece is the next odd number.



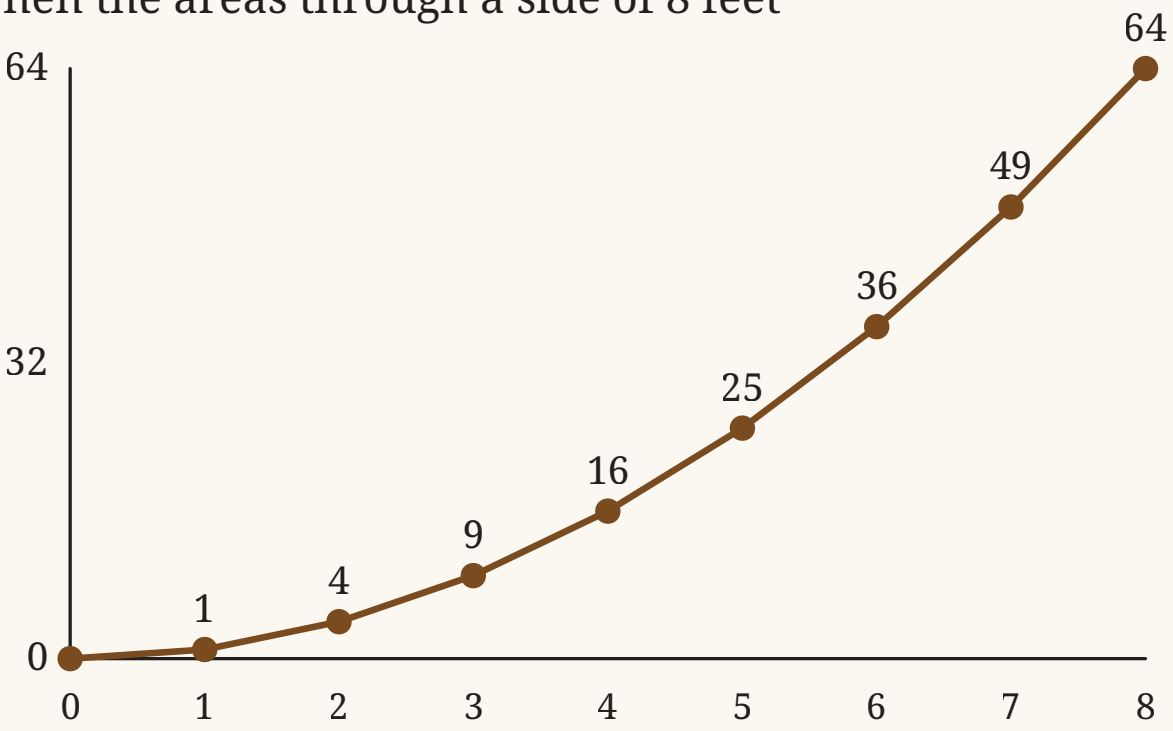
1

4, and the new piece is 3

9, and the new piece is 5

16, and the new piece is 7

Then the areas through a side of 8 feet




Vertical scale: square feet, for this picture only. It starts at 0.

The square grows by odd pieces. Sides 1, 2, 3, and 4 have areas 1, 4, 9, and 16. The new pieces are 3, then 5, then 7. The curve then marks 1, 4, 9, 16, 25, 36, 49, and 64 square feet. That vertical scale belongs to this picture only.

| Side, feet | Area, square feet |
|---|---|
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |
| 4 | 16 |
| 5 | 25 |
| 6 | 36 |
| 7 | 49 |
| 8 | 64 |

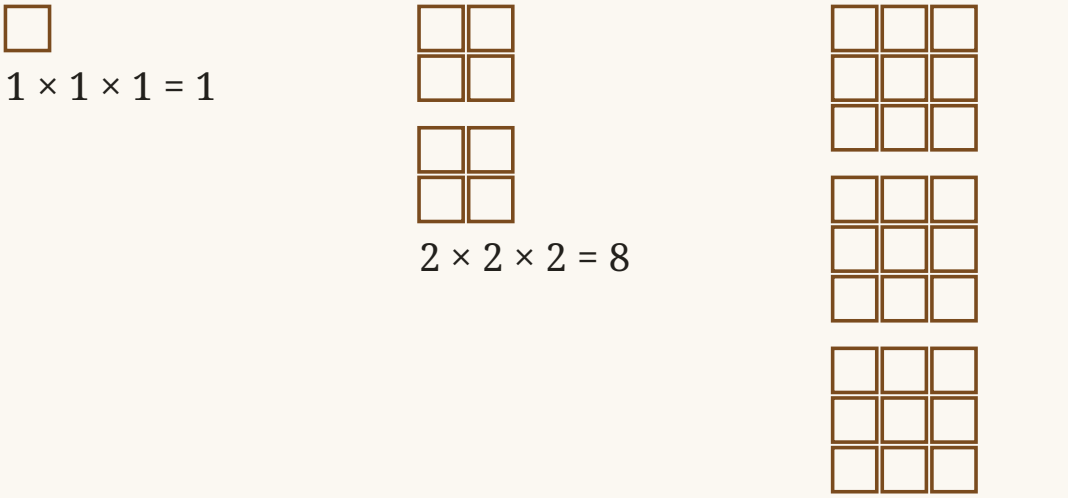
The hurricane page uses the same square on wind speed, with the air and the wall held still.

## Cubic: the cube

Cubic growth follows the volume of a cube. A cubic foot is a cube one foot on each side. A concrete cube 2 feet on a side has 2 times 2 times 2, which is 8 cubic feet.

# A cube counts side, three times

Each small square is one cubic foot in that layer.

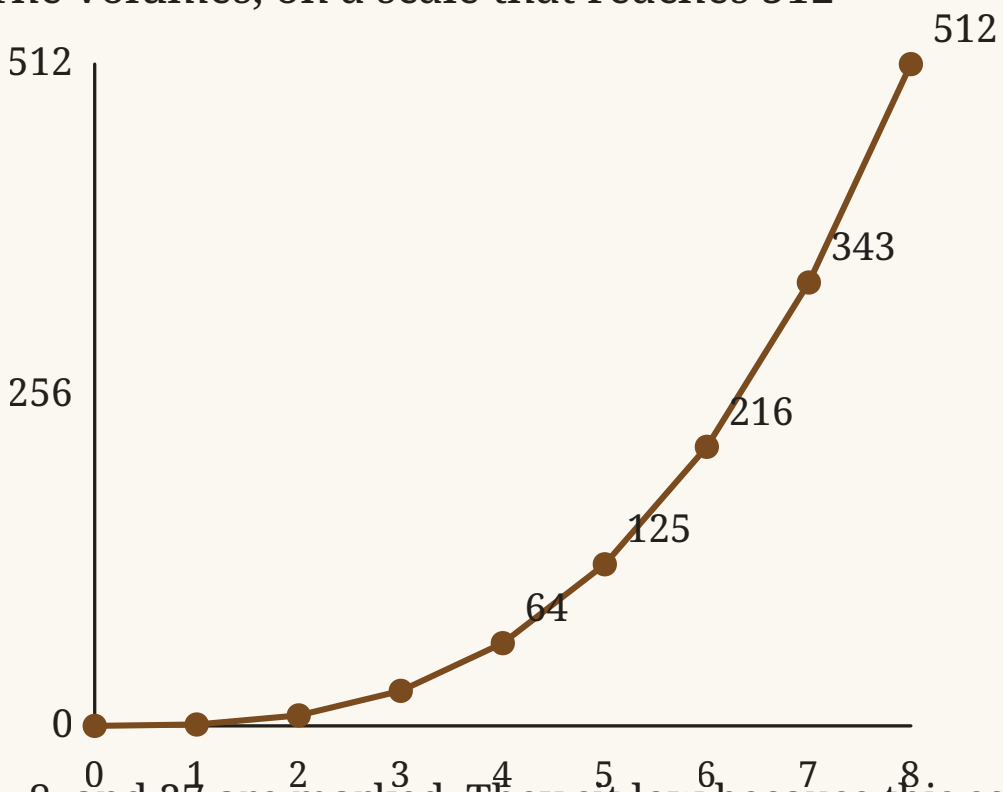


1 × 1 × 1 = 1

2 × 2 × 2 = 8

3 × 3 × 3 = 27

The volumes, on a scale that reaches 512




1, 8, and 27 are marked. They sit low because this scale reaches 5

Cubic feet, this picture only. The drawing counts unit cubes.

A cube counts the side three times. The layers show 1, then 8, then 27 cubic feet. The curve marks 1, 8, 27, 64, 125, 216, 343, and 512. The scale reaches 512, so 1, 8, and 27 sit low. Cubic feet, this picture only.

| Side, feet | Volume, cubic feet |
|---|---|
| 1 | 1 |
| 2 | 8 |
| 3 | 27 |
| 4 | 64 |
| 5 | 125 |
| 6 | 216 |
| 7 | 343 |
| 8 | 512 |

## Exponential: multiply each time

Exponential growth multiplies by the same factor at each equal step. Imagine bacteria doubling each hour, starting from one at hour zero. The table starts at hour 1.

| Hours | Ideal count |
|---|---|
| 1 | 2 |
| 2 | 4 |
| 3 | 8 |
| 4 | 16 |
| 5 | 32 |
| 6 | 64 |
| 7 | 128 |
| 8 | 256 |

Each line is twice the line before. Bacteria cannot double forever. The model assumes unlimited food and space. Early on, doubling can sit below the square. Later it passes the square, because a fixed multiplier, repeated, eventually outruns adding larger pieces. At step 3, doubling gives 8 and the square gives 9. At step 4 both give 16. At step 5, doubling gives 32 and the square gives 25. Those are output numbers of the two rules, not miles set against a count of bacteria.

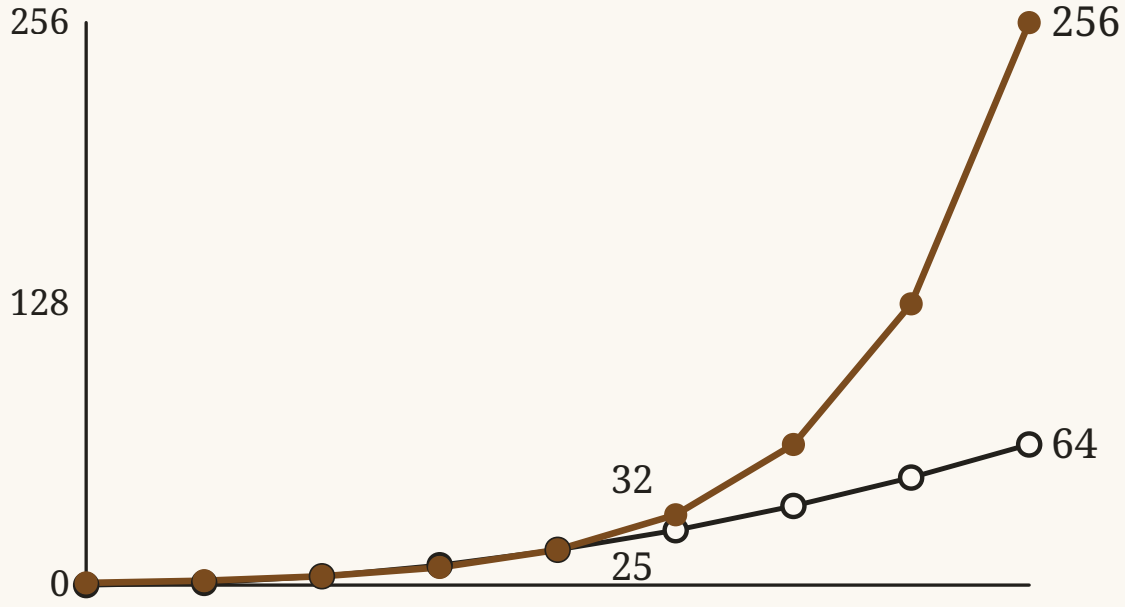
# Each step multiplies by 2

Start with 1 at hour 0. The table begins at hour 1, which is 2.




Then ×2 keeps going: 32, 64, 128, 256.

The output numbers, on one scale from 0 to 256
Square feet and a bacteria count are different things.



Each step multiplies by 2. From 1 at hour 0 the counts are 2, 4, 8, 16, 32, 64, 128, and 256. On one scale from 0 to 256, the square's numbers 1, 4, 9, 16, 25, 36, 49, and 64 are drawn beside them. At step 3, 8 is below 9. At step 5, 32 is above 25.

## Factorial: count the orders

A factorial counts the different ways to seat distinct people in a row. The sign ! is the name of that count, not a mark of excitement. For a group of n people, n! multiplies the whole numbers from 1 through n. Three people can sit in a row of three seats in 3 times 2 times 1, which is 6 ways.

# One more person multiplies

The sign ! names the count. It is not excitement.

| | | |
|---|---|---|
| 1 person | A | 1 way |
| 2 people | AB BA | 2 ways |
| 3 people | ABC ACB BAC<br>BCA CAB CBA | 6 ways |

After three, the list is a multiplication.

| | | |
|---|---|---|
| 1 person | | 1 way |
| 2 people | × 2 | 2 ways |
| 3 people | × 3 | 6 ways |
| 4 people | × 4 | 24 ways |
| 5 people | × 5 | 120 ways |
| 6 people | × 6 | 720 ways |
| 7 people | × 7 | 5,040 ways |
| 8 people | × 8 | 40,320 ways |

These rows are equally spaced so the multiplier can be read. They are not bars sized to the count. 40,320 would hide the 6.

One more person multiplies the arrangements. One person: 1 way. Two people: AB and BA, 2 ways. Three people: ABC, ACB, BAC, BCA, CAB, and CBA, 6 ways. Then times 4 is 24, times 5 is 120, times 6 is 720, times 7 is 5,040, and times 8 is 40,320. The rows are equally spaced so the multiplier can be read. They are not bars sized to the count.

| People n | Arrangements n! |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 6 |
| 4 | 24 |
| 5 | 120 |
| 6 | 720 |
| 7 | 5,040 |
| 8 | 40,320 |

For these fixed rules, the factorial eventually grows fastest. Rankings among small inputs are different. Do not promote a small-input race into a law.

## Logarithmic: count the multiplications

A logarithm adds a fixed amount when the input grows by a fixed factor. log10 counts tenfold steps from 1. Each step multiplies by ten. The inputs here are powers of ten, not the numbers 1 through 8. "Digits" means how many figures you write. 100 has three.

# Times ten, add one step

Each row is ten times the input above it.

| Input | Steps | Digits |
|---|---|---|
| 1 | 0 | 1 |
| ×10 | | |
| 10 | 1 | 2 |
| ×10 | | |
| 100 | 2 | 3 |
| ×10 | | |
| 1,000 | 3 | 4 |
| ×10 | | |
| 10,000 | 4 | 5 |
| ×10 | | |
| 100,000 | 5 | 6 |
| ×10 | | |
| 1,000,000 | 6 | 7 |
| ×10 | | |
| 10,000,000 | 7 | 8 |

Equal row gaps mean ×10, not ordinary counting gaps.
Digits are the steps plus one, and they jump by whole numbers.

Times ten, add one step. The inputs are 1, 10, 100, 1,000, 10,000, 100,000, 1,000,000, and 10,000,000. The tenfold steps are 0, 1, 2, 3, 4, 5, 6, and 7. The written digits are 1 through 8. Equal row gaps mean each input is ten times the one before, not an ordinary counting gap. Digits are the steps plus one.

| Input | Tenfold steps, log10 | Written digits |
|---|---|---|
| 1 | 0 | 1 |
| 10 | 1 | 2 |
| 100 | 2 | 3 |
| 1,000 | 3 | 4 |
| 10,000 | 4 | 5 |
| 100,000 | 5 | 6 |
| 1,000,000 | 6 | 7 |
| 10,000,000 | 7 | 8 |

Digit counts jump by whole numbers. They are not a smooth curve. For these powers of ten, log10 plus one matches the digit count.

The sheet is Growth rates compared (PDF).

## Key facts

- Linear adds a constant. The odometer example adds 10 miles each hour.
- Quadratic outputs are 1, 4, 9, 16, and so on. Cubic outputs are 1, 8, 27, 64, and so on.
- Exponential doubling gives 2, 4, 8, 16. The model assumes unlimited resources.
- 8! is 40,320. log10 of 1,000,000 is 6, and that number is written with 7 digits.